

RealMan Robotic Arm rm_moveit2 User Manual V1.0

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Revision History

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rm_moveit2_Package_Description

rm_moveit2 is a Moveit2 control example package for RealMan robotic arms. It currently provides launch entries for ECO63, RM75, and RX75 dual-arm motion planning demos.

- 1.Package use.
- 2.Package architecture description.

Through the introduction of these two parts, it can help you:

- 1.Understand the package use.
- 2.Familiar with the file structure and function of the package.

rm_moveit2_Package_Use

Basic_Package_Use

Before running rm_moveit2, first start the corresponding rm_driver, rm_description, rm_control, and Moveit configuration package nodes. For ECO63 and RM75, the Moveit environment can be started with the corresponding rm__config package. For RX75 dual-arm, please use the dedicated rm_rx75_config package.

Start the ECO63 example with the following command.

```
rm@rm-desktop:~$ ros2 launch rm_moveit2 moveit_eco63.launch.py
```

Start the RM75 example with the following command.

```
rm@rm-desktop:~$ ros2 launch rm_moveit2 moveit_rm75.launch.py
```

Start the RX75 dual-arm example with the following command.

```
rm@rm-desktop:~$ ros2 launch rm_moveit2 moveit_rx75.launch.py
```

Before starting the RX75 dual-arm example, use the following command to launch the corresponding Moveit environment.

```
rm@rm-desktop:~$ ros2 launch rm_rx75_config demo_6fb_v.launch.py
```

Advanced_Package_Use

The common parameters of rm_moveit2 are configured in the launch files. Parameter `planning_group`: the planning group used by Moveit. The default value is `rm_group` for ECO63 and RM75, and `right_arm` for RX75. Parameter `current_state_wait_sec`: the waiting time for reading the current state. Parameter `velocity_scaling`: the motion velocity scaling factor. Parameter `acceleration_scaling`: the motion acceleration scaling factor. Parameter `planning_time`: the planning time of Moveit. Parameter `home_named_target`: the named target used when moving to the initial posture. Parameter `enable_pose_target`: whether to directly plan to the pose target given by `pose_target_csv`. Parameter `pose_target_csv`: the pose target in `x,y,z,rx,ry,rz` format. Parameter `pose_reference_frame`: the reference frame used for the pose target. Parameter `prefer_named_start`: this parameter is only used by the RX75 dual-arm launch file to prefer the named start posture. Parameter

enable_cartesian_demo: this parameter is only used by the RX75 dual-arm launch file to control whether the Cartesian demo is executed. For the RX75 dual-arm example, planning_group can be set to left_arm or right_arm according to the selected arm side.

rm_moveit2_Package_Architecture_Description

Overview_of_Package_Files

The current rm_moveit2 package is composed of the following files.

```
├── CMakeLists.txt
├── launch
│   ├── moveit_eco63.launch.py          # ECO63 launch file
│   ├── moveit_rm75.launch.py          # RM75 launch file
│   └── moveit_rx75.launch.py          # RX75 dual-arm launch file
├── package.xml
├── README.md
└── src
    ├── linear_motion_node.cpp          # MoveIt motion helper file
    └── simple_moveit_node.cpp          # main node file
```